BITS Pilani

Introduction to Statistical Methods

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REC

Session 12
(25th January 2026)

REC

Prediction and Forecasting

Contact Session 12

Contact Session 12: : Module 5: Prediction & Forecasting

Contact Session	List of Topic Title	Reference
CS - 12	Time Series Analysis - Introduction, Components of time series data, Moving Averages and weighted moving averges model	T3
HW	Problems on Time series	T3

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Case Study

Scenario 1: A teacher collects **study hours and marks obtained** by students data.

Study Hours (X)	Marks (Y)
2	40
4	55
6	70
8	85

Scenario 2: A college records **number of student admissions every year**.

Year	Admissions
2020	450
2021	480
2022	520
2023	580
2024	610

- Examine the two datasets provided. Explain how they differ and decide which statistical method should be used to analyze each dataset.

Correlation Vs Time Series

	Correlation	Time Series
Type of Data	Two or more variables measured at the same time.	One variable measured at different times.
Time Dependency	No time factor involved. Time is irrelevant to the calculation.	Time is compulsory. The entire analysis relies on the chronological sequence.

Correlation Vs Time Series

Basis	Correlation	Time Series
Meaning	Measures relationship between two variables	Studies data collected over time
Purpose	Find strength of relation	Pattern, trend, prediction
Data	2+ variables, same time	1 variable, different times
Dependency	No time factor involved	Time is compulsory
Output	Coefficient (r) between −1 and +1	Trend, seasonal, cyclical components
Used For	Relationship analysis	Forecasting

Agenda

- Introduction to Time Series
- Components of Time Series
- Moving Average Model
- Weighted Moving Average Model

Introduction to Time Series

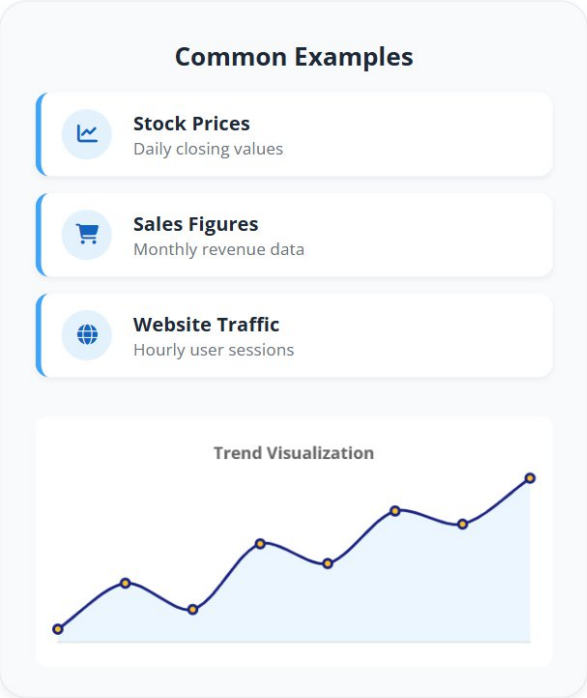
- Definition: A sequence of data points or observations measured at successive, equally spaced intervals of time (t = 1, 2, …). It focuses on understanding how a variable changes over time.

Key Terms

- Observation (Y_t)
- Time Index (t)
- Lag (k)
- Frequency (e.g., Daily)

Primary Goals

- **Describe:** Identify patterns like trends, cycles, and seasonality.
- **Smooth:** Filter out random noise to reveal the underlying signal.
- **Forecast:** Predict future values based on historical patterns.



Introduction to Time Series

- A time series is defined as a set of observations on a variable generated sequentially in time. The measurement of the variable are made at discrete (equally spaced) intervals.
- Often a variable continuous in time is measured at regular intervals and this produces a discrete series of data. Such data may be collected at regular time intervals such as, monthly, weekly, quarterly or annually.
- Time series are used in statistics, econometrics, mathematical finance, weather forecasting, earthquake prediction and many other applications.

Introduction to Time Series

- Time series may be represented as a set of data (Y_1, Y_2, Y_3, …, Y_n), Y_t denoting the value of the variable Y at time t.
- A discrete time series data may be generated from the accumulation of data over a period of time.

Example of continuous variable:

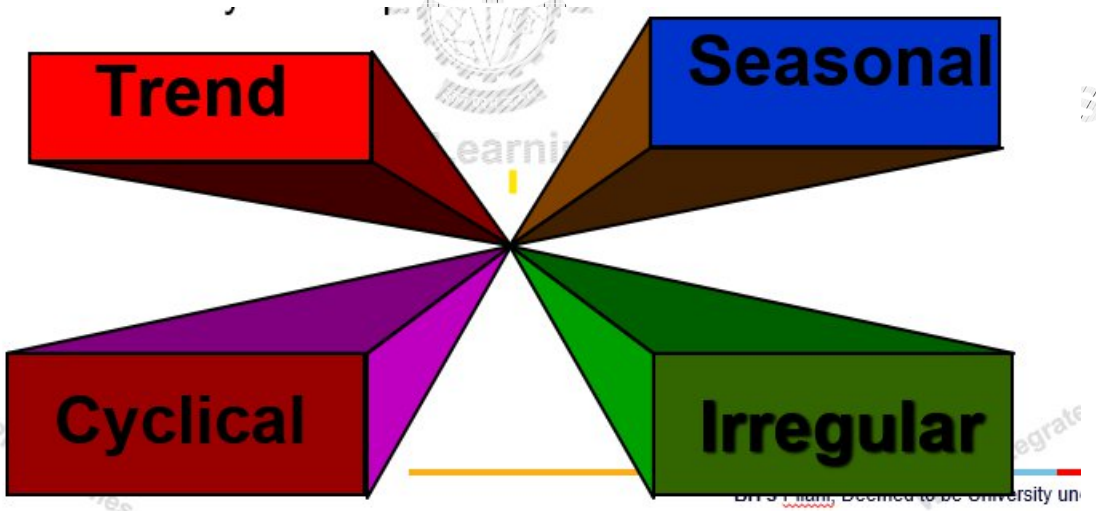
- Temperature on a chemical reactor
- Level of tide at a particular site
- Amplitude of an electrical signal

Example of discrete variable:

- Monthly sales
- Daily rainfall
- Production of a crop over different years in a country

Decomposition of Time Series

- Decomposition is an important technique for all types of time series analysis.
- It is used to reconstruct the original time series by additions or multiplications of component where each of these has a certain characteristic or type of behaviour.
- Time series are usually decomposed into:



Decomposition/Components of Time Series

Trend T

Long-term increase or decrease in the data level over time. Indicates the general direction of the series.

Seasonality S

Short-term regular patterns repeating over a fixed period (e.g., daily, weekly, quarterly) driven by calendar factors.

Cyclical C

Long-term wave-like fluctuations tied to broader economic or business cycles. Irregular duration unlike seasonality.

Irregular / Noise I

Random, unpredictable residual variation that remains after extracting other components. Contains no pattern.

ADDITIVE MODEL

$$Y_t = T + S + C + I$$

MULTIPLICATIVE MODEL

$$Y_t = T \times S \times C \times I$$

Act, 1956

Trend component

- T_t: the trend component at time *t*, which reflects the long-term progression of the series (secular variation).
- A trend exists when there is a persistent increasing or decreasing direction in the data. **The trend component does not have to be linear**

- Trend measure the change in the mean level of the time series.
- Persistent, overall upward or downward pattern
- Several years duration
- Indicates long term movements in the level of the series.

Mo., Qtr., Yr.

Seasonal component

- S_t (seasonality): the seasonal component at time *t*.
- A seasonal pattern exists when a time series is influenced by seasonal factors. Seasonality occurs over a fixed and known period (e.g., the quarter of the year, the month, or day of the week).
- Generally within a year or shorter we get seasonal component in the series.

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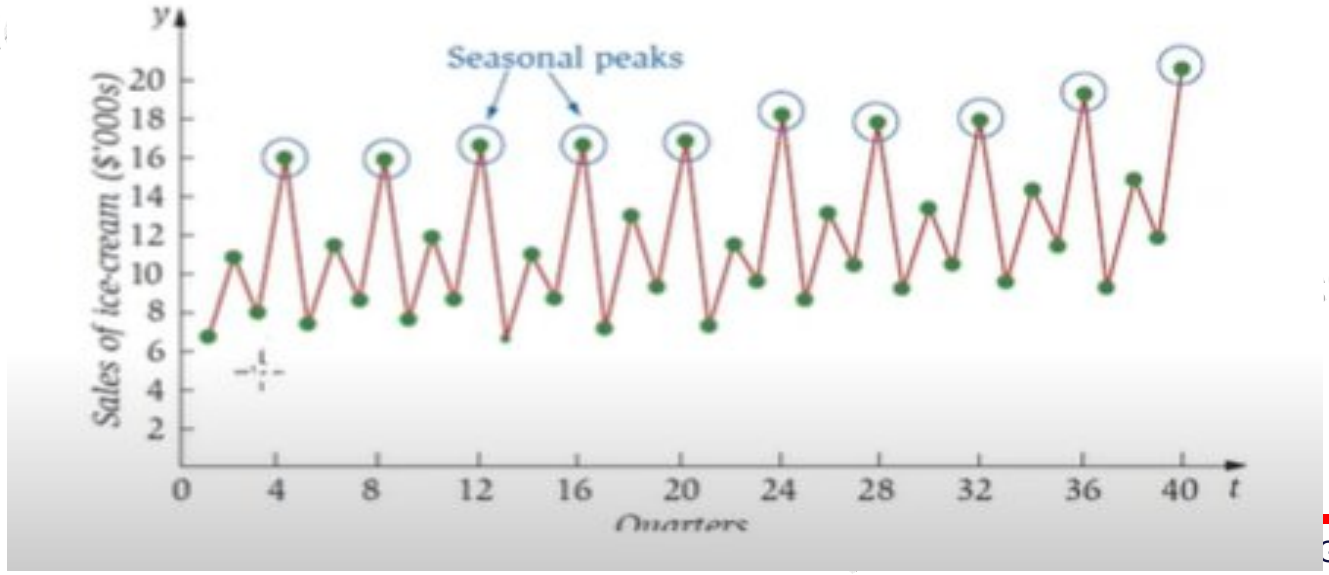
- The seasonal variation of a time series is a pattern of change that **recurs** regularly over time.
- Seasonal variations are usually due to the differences between seasons and to festive occasions such as Hindu, Christian or Muslim festivals.

❖ Examples include:

- Air conditioner sales in Summer
- Heater sales in Winter
- Flu cases in Winter
- Airline tickets for flights during school vacations

Seasonal Component

- Seasonality in a time series can be identified by **regularly spaced peaks and troughs** which have a consistent direction and approximately the same magnitude every year, relative to the trend.
- The following diagram depicts a strongly seasonal series. There is an obvious large seasonal increase in December retail sales in New South Wales due to Christmas shopping. In this example, the magnitude of the seasonal component increases over time, as does the trend.



GC Act, 1956

Cyclic component

- C_t: the cyclical component at time *t*.
  which reflects repeated but non-periodic fluctuations.
  The duration of these fluctuations are extended over a long period.
  Usually are longer than a year.

- Cyclical variations also have recurring patterns but with a longer and more **erratic time scale** compared to Seasonal variations.

- The name is quite misleading because these cycles can be far from regular and it is usually impossible to predict just how long periods of expansion or contraction will be.

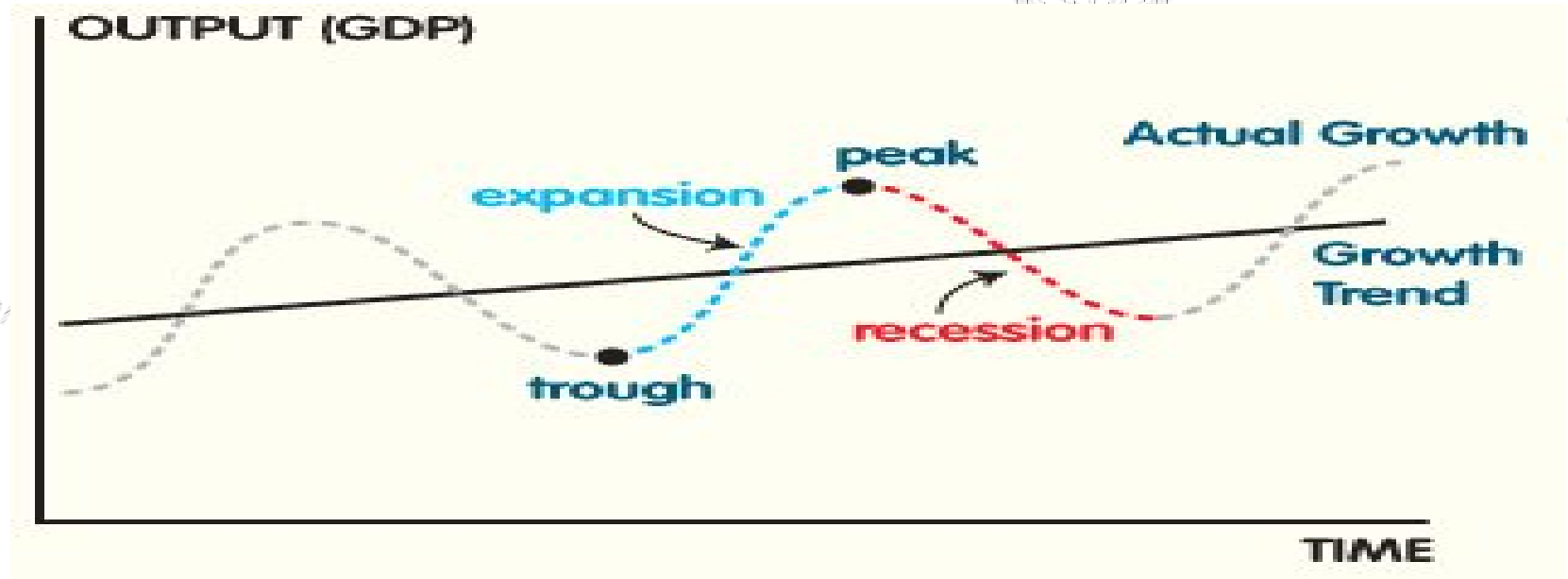
- There is no guarantee of a regularly returning pattern.

Cyclic component

Examples:

- Floods
- Wars
- Changes in interest rates
- Economic depressions or recessions
- Changes in consumer spending

This chart represents an economic cycle, but we know it doesn't always go like this. The timing and length of each phase is not predictable.



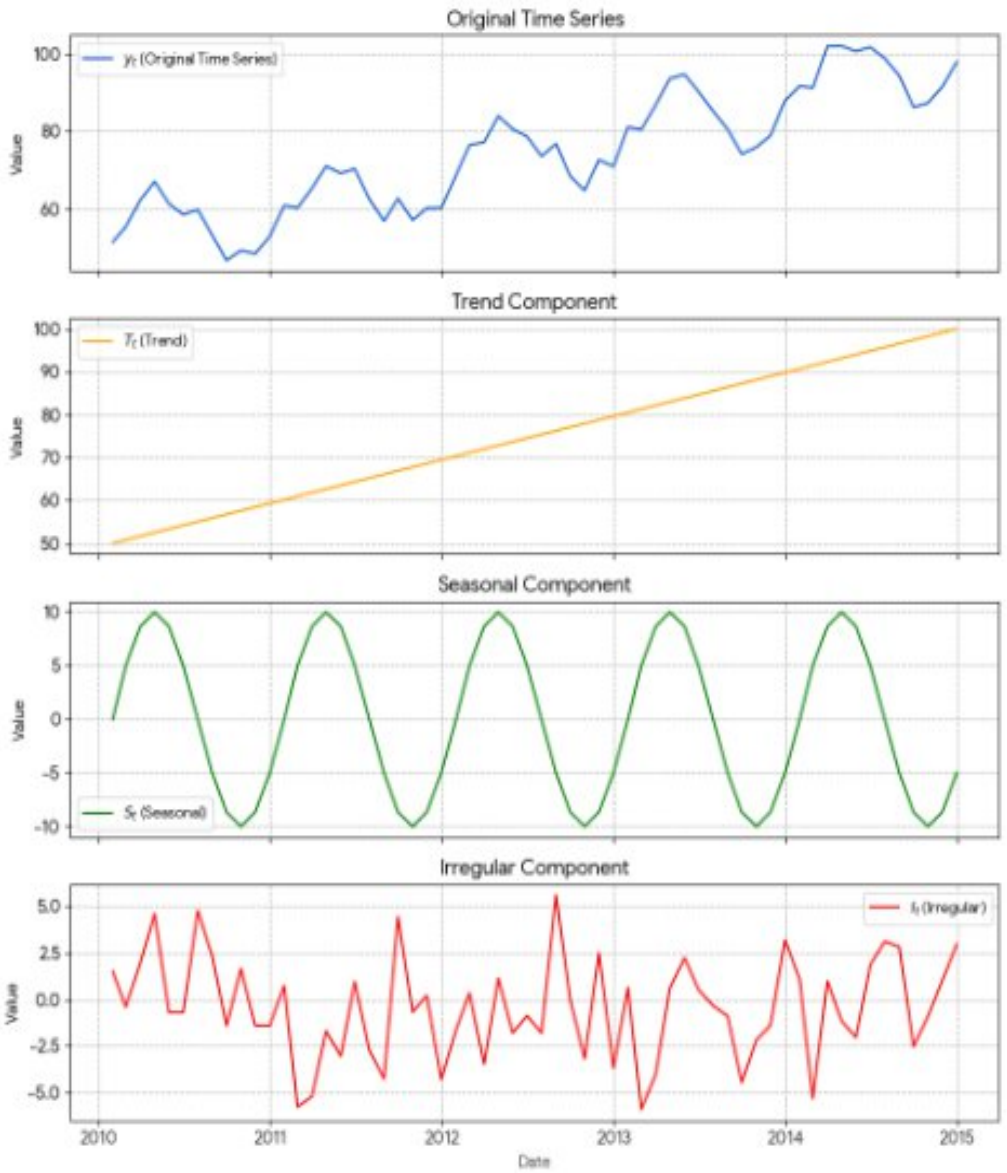
Difference Between Seasonal and Cyclic Components

Feature	Seasonal Component	Cyclic Component
Definition	A repeating pattern that occurs at fixed, regular intervals (e.g., daily, monthly, yearly).	A fluctuating pattern that occurs at irregular intervals and is influenced by external factors like economic cycles.
Periodicity	Fixed and known (e.g., quarterly, annual, weekly).	Variable and unpredictable (can last several years).
Cause	Usually driven by natural, social, or institutional factors (e.g., weather, holidays).	Driven by economic, political, or business cycles.
Duration	Short-term and repeats within a fixed period.	Long-term and may not have a fixed period.
Examples	- Higher ice cream sales in summer. - Increased shopping in December (holiday season).	- Stock market fluctuations. - Economic recessions and expansions.

Irregular component

- I_t: the irregular component (or "noise") at time *t*.
- ❖ which describes random or short term unpredictable fluctuations.

- It represents the residuals or remainder of the time series after the other components have been removed.



Irregular component

- An irregular (or random) variation in a time series occurs over varying (usually short) periods.

- It follows no pattern and is by nature unpredictable.
- It usually occurs randomly and may be linked to events that also occur randomly.
- Irregular variation cannot be explained mathematically.

- If the variation cannot be accounted for by secular trend, season or cyclical variation, then it is usually attributed to irregular variation. Examples:
  - Sudden changes in interest rates
  - Collapse of companies
  - Natural disasters
  - Sudden shift s in government policy
  - Dramatic changes to the stock market
  - Effect of Middle East unrest on petrol prices.

Irregular component

- Erratic, unsystematic, ‘residual’ fluctuations
- Due to random variation or unforeseen events like Union strike, War Short duration & nonrepeating
- Rapid changes or bleeps in the data caused by short term unanticipated and non-recurring factors. Irregular fluctuations can happen as often as day to day

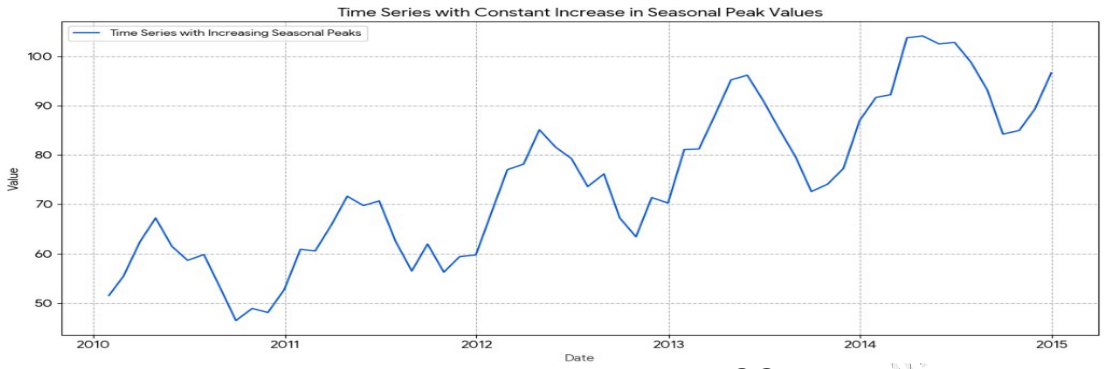
- The irregular component results from short term fluctuations in the series which are neither systematic nor predictable. In a highly irregular series, these fluctuations can dominate movements, which will mask the trend and seasonality.

Additive model

$$y_t = T_t + C_t + S_t + I_t$$

Characteristics:

- The seasonal variations remain constant over time.
- Suitable when seasonal fluctuations are of roughly equal magnitude, regardless of the trend level.
- Used when data exhibits linear patterns.
- **Example:**
- If ice cream sales increase by 100 units every summer regardless of the overall trend, an additive model is appropriate.



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Multiplicative model

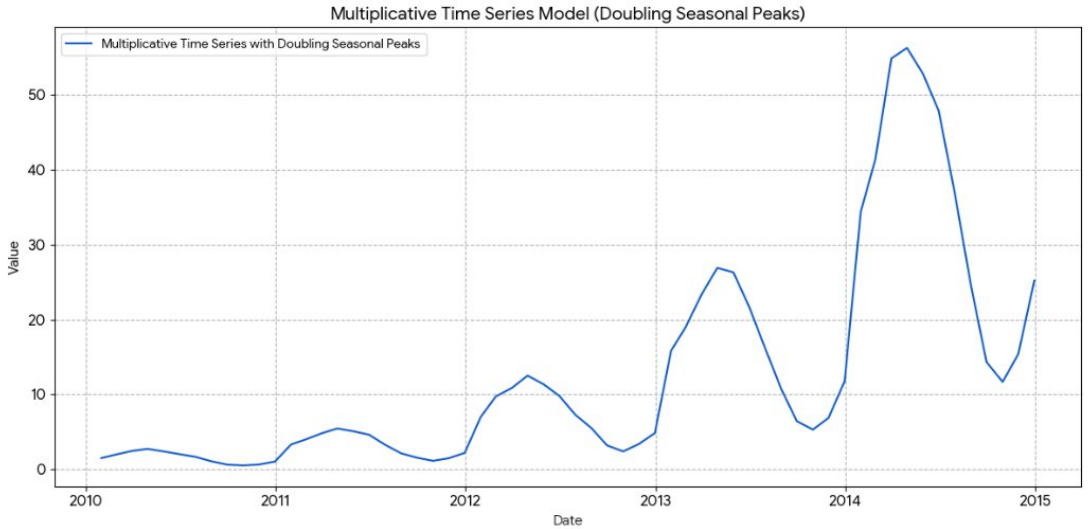
$y_t = T_t \times C_t \times S_t \times I_t$ or equivalently, $\log(y_t)=\log(T_t)+\log(C_t)+\log(S_t)+\log(I_t)$ (after taking logarithm, it transforms into an additive form).

Characteristics:

- The seasonal variations change in proportion to the trend (i.e., they grow or shrink as the trend changes).
- Suitable for data where seasonal effects increase or decrease with the level of the series.
- Used when data exhibits exponential growth patterns.

Example:

If ice cream sales double every summer relative to the previous year's sales, a multiplicative model is more suitable.



Key Differences

Feature	Additive Model	Multiplicative Model
Formula	$y_t = T_t + C_t + S_t + I_t$	$y_t = T_t \times C_t \times S_t \times I_t$
Seasonality	Constant magnitude	Varies with trend (proportional)
Trend	Linear	Exponential or non-linear
When to Use	Seasonal variations are stable	Seasonal variations change with trend
Graph Behavior	Seasonal pattern remains constant	Seasonal amplitude increases/decreases over time

Choosing Between the Two:

- Use the **additive model** when seasonal fluctuations are independent of trend magnitude.
- Use the **multiplicative model** when seasonal variations increase or decrease as the trend changes.
- If unsure, visualize the data: if seasonal patterns appear to grow with time, the multiplicative model is likely the better choice.

Difference Between Forecasting and Prediction

Criteria	Forecasting	Prediction
Definition	○ Estimating future events based on past data, patterns, and trends using statistical models or quantitative analysis.	○ Making judgments or guesses about future events, often without relying heavily on historical data or models.

Forecasting

- **Forecasting** is the process of making **predictions of the future** based on **past and present data** and most commonly by analysis of trends.

❖ Forecasts can be required several years in advance.
❖ Forecasting is required in many situations:

(a) deciding whether to build another power generation plant in the next five years requires forecasts of future demand.
(b) scheduling staff in a call centre next week requires forecasts of call volumes
(c) stocking an inventory requires forecasts of stock requirements. etc.

- Whatever the circumstances or time horizons involved, forecasting is an important aid to effective and efficient planning.

Forecasting

- Some things are easier to forecast than others. The time of the sunrise tomorrow morning can be forecast precisely.
- On the other hand, tomorrow's iPhone sale cannot be forecast with any accuracy.

Forecasting depends on several factors including:

- how well we understand the factors that contribute to it;
- how much data is available;
- whether the forecasts can affect the thing we are trying to forecast.

- Forecasts of the time of the sunrise tomorrow morning, electricity demand can be highly accurate because all three conditions are usually satisfied.

- Forecasting currency exchange rates, only one of the conditions is satisfied: there is plenty of available data. However, we have a limited understanding of the factors that affect exchange rates.

Forecasting, Planning and Goals

- ***Forecasting*:**
is about predicting the future as accurately as possible, given all of the information available, including historical data and knowledge of any future events that might impact the forecasts

***Planning*:**
is a response to forecasts and goals. Planning involves determining the appropriate actions that are required to make your forecasts match your goals.

- ***Goals*:**
are what you would like to have happen. Goals should be linked to forecasts and plans, but this does not always occur. Too often, goals are set without any plan for how to achieve them, and no forecasts for whether they are realistic.

Forecasting frame

Short-term Forecasts

- Short-term forecasts generally involve some form of scheduling which may include for example the seasons of the year for planning purposes.
- The cyclical and seasonal factors are not more important in these situations.
- Such forecasts are usually prepared every 6 months or on a more frequent basis.
- These forecasts are needed for the scheduling of personnel, production and transportation. As part of the scheduling process, forecasts of demand are often also required.

Forecasting frame

Medium-term Forecasts

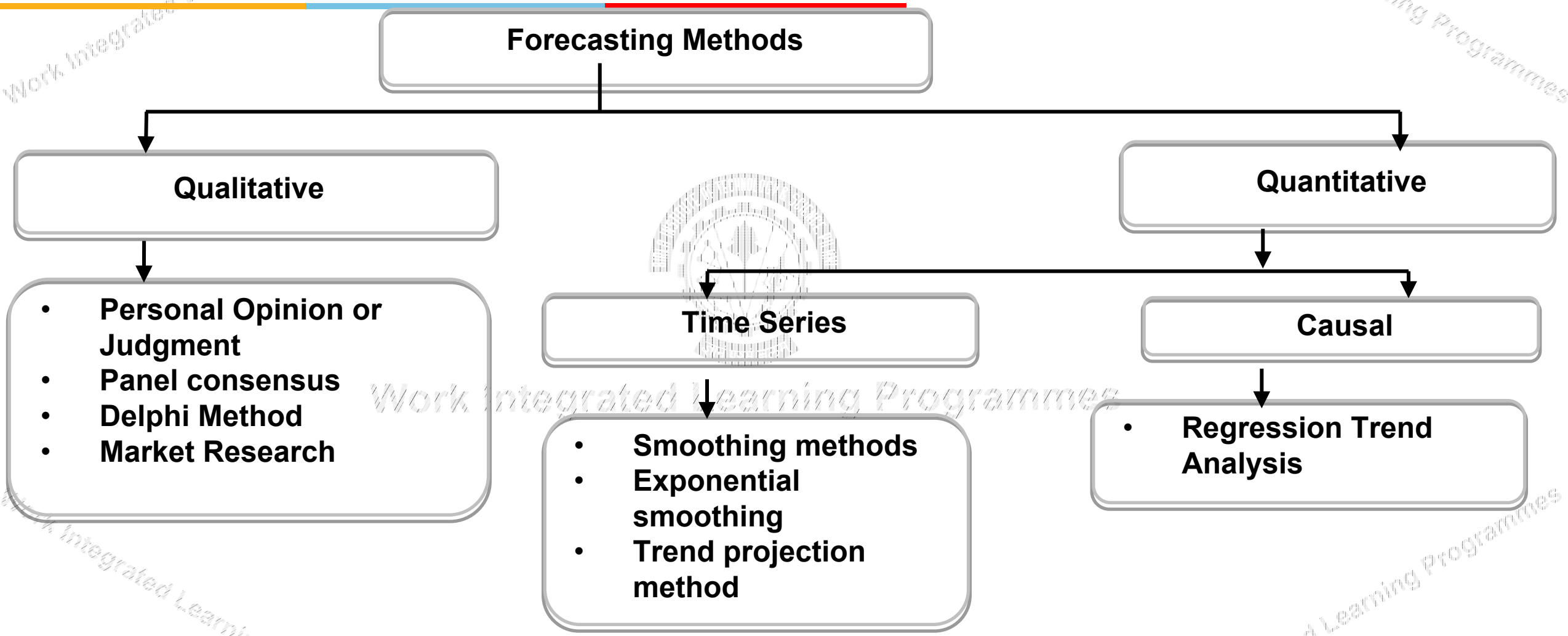
- Medium-term forecasts are generally prepared for planning, scheduling, budgeting and resource requirements purposes.
- Medium-term forecasts are needed to determine future resource requirements, in order to purchase raw materials, hire personnel, or buy machinery and equipment.
- The trend factor, as well as the cyclical component, plays a key role in the medium-term forecast as the year to year variations in traffic growth are an important element in the planning process

Forecasting frame

Long-term Forecasts

- Long-term forecasts are used mostly in connection with strategic planning to determine the level and direction of capital expenditures and to decide on ways in which goals can be accomplished.
- The trend element generally dominates long term situations and must be considered in the determination of any long-run decisions.
- The methods generally found to be most appropriate in long-term situations are econometric analysis and life-cycle analysis. Long-term forecasts are used in strategic planning. Such decisions must take account of market opportunities, environmental factors and internal resources.
- It is also important that since the time span of the forecast horizon is long, forecasts should be calibrated and revised at periodic intervals (every two or three years depending on the situation).

Forecasting Methods



Smoothing Techniques – Moving Averages

- **Appropriate for** a time series with a horizontal pattern ie., **the data that are stationary.**
  - Moving Average (the average of the most recent k data values forms the forecast for the next period)

$$F_{t+1} = \frac{\sum \text{most recent k data values}}{k}$$

$$F_{t+1} = \frac{Y_t + Y_{t-1} + \ldots + Y_{t-k+1}}{k}$$

Measurement of forecast error

What was forecasted to happen at time t using the forecast technique

Forecast Error at time t

$$\text{Forecast Error}_t = y_t - F_t$$

What actually happened at time t

Performance Measures

Mean Square Error (MSE)

$$MSE = \frac{\sum (\Delta_t)^2}{n}$$

Mean Absolute Percent Error (MAPE)

$$MAPE = \frac{\sum \frac{|\Delta_t|}{y_t}}{n} 100$$

Mean Absolute Deviation (MAD)

$$MAD = \frac{\sum |\Delta_t|}{n}$$

Largest Absolute Deviation (LAD)

$$LAD = Max\,|\Delta_t|$$

Choice of the Performance Measure

Which performance should be used?

Advantages

- MSE gives greater weight to larger deviations (which could result from outliers)

$$MSE = \frac{\sum(\Delta_t)^2}{n}$$

- MAD gives less weight to larger deviations

$$MAD = \frac{\sum|\Delta_t|}{n}$$

- MAPE gives less overall weight to a large deviation if the time series value is

$$MAPE = \frac{\sum \frac{|\Delta_t|}{y_t}}{n} 100$$

- LAD tells us if all deviations fall below some threshold value

$$LAD = Max\,|\Delta_t|$$

Although we illustrate all 4 techniques, in general focus will be on MSE and MAD. A

Simple Moving Average

- The idea behind the moving averages is that observations which are nearby in time are also likely to be close in value.
- The average of the points near an observation will provide a reasonable estimate of the trend-cycle at that observation.
- The average eliminate some of the randomness in the data, and leaves a smooth trend-cycle component.

- It is assumed that fluctuations are around some base(level) line.
- If we take more data for average then we get more smooth curve.
- Thus method require more data for better forecast.
- If we take one year moving average, it's a naive forecast.

Simple Moving Average

SIMPLE MOVING AVERAGE (SMA)

Calculates the unweighted mean of the previous **k** data points.

$$MA_t = \frac{Y_t + Y_{t-1} + \ldots + Y_{t-k+1}}{k}$$

Methodology

Choose Window (k)

Larger k: Smoother, less noise, slow reaction.
Smaller k: More responsive, follows noise.

Forecast Rule

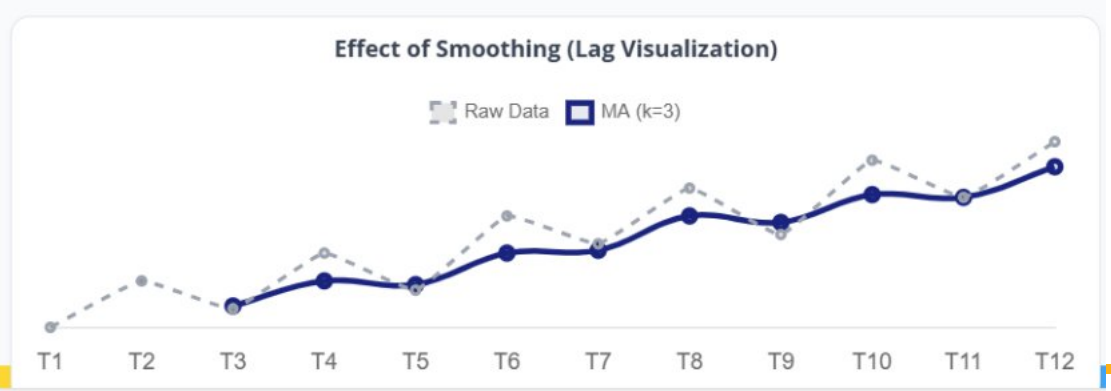
The forecast for the next period is simply the current moving average value:
$F_{t+1} = MA_t$

Pros

- Easy to calculate and understand.
- Effective at smoothing random noise.
- Robust to occasional outliers.

Cons

- Lags behind trends (delayed reaction).
- Treats all past k observations equally.
- Cannot predict complex seasonality alone.



Common Use Cases

- Inventory Control
- Demand Planning
- Sales Baseline
- Process Monitoring

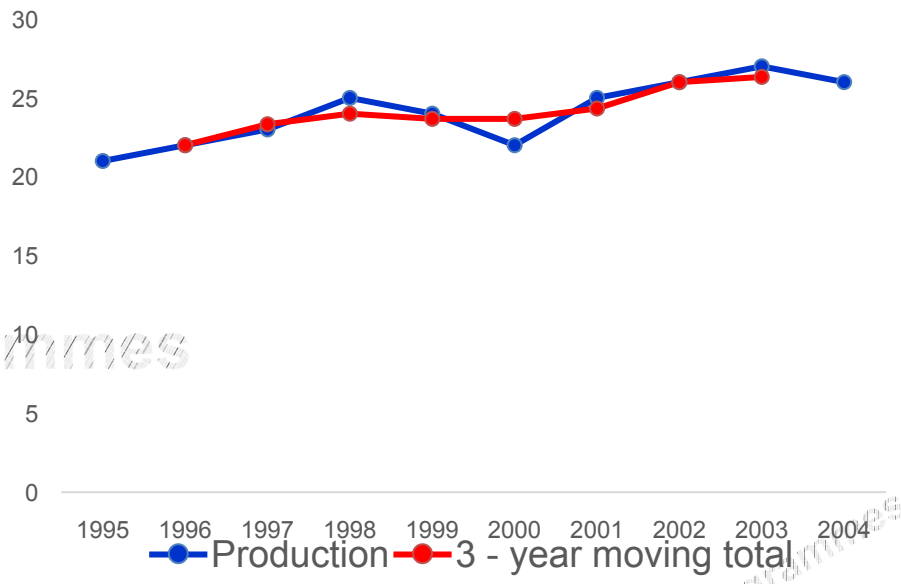
Key Differences

Feature	Moving Average (MA)	Centered Moving Average (CMA)
Definition	A method that averages past data points over a fixed window to smooth fluctuations.	A refined version of MA that aligns the smoothed values to the center of the window.
Formula		
Positioning of Values	The average is placed at the end of the window.	The average is placed at the center of the window.
Use Case	**Used for short-term forecasting and smoothing noisy data.**	**Used primarily for smoothing the time series.**
Effect on Trend	Introduces a lag since it is calculated using past values.	Better aligns with the actual values by placing the average in the correct time period.
Window Type	Uses past values only (trailing moving average).	Uses both past and future values (symmetric window).
Example		4-period CMA: Calculate 2-period MA first, then average adjacent MA values to align to the center.

Example 1

Following data shows production volume (in '000 tones). Compute 3-year centre moving average for all available years

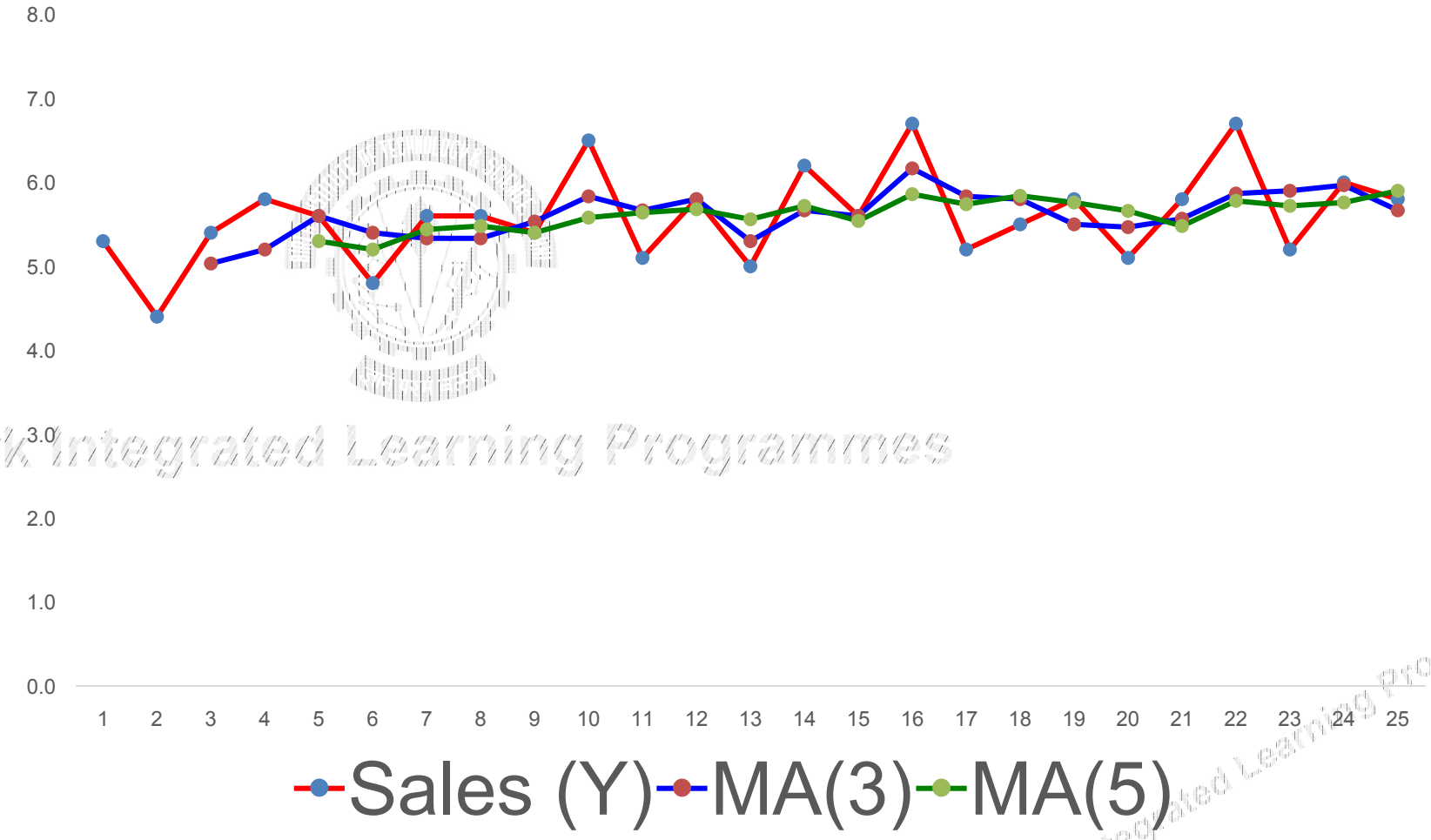
Year	Production	3 - year moving total	3 - year moving average
1995	21		
1996	22	66	22.00
1997	23	70	23.33
1998	25	72	24.00
1999	24	71	23.67
2000	22	71	23.67
2001	25	73	24.33
2002	26	78	26.00
2003	27	79	26.33
2004	26		



Example 2

Following data shows production volume (in '000 tones). Compute 3 and 5-year moving average

Period (t)	Sales (Y_t)	3yr MA	5yr MA
1	5.3		
2	4.4	5.03	
3	5.4	5.20	5.30
4	5.8	5.60	5.20
5	5.6	5.40	5.44
6	4.8	5.33	5.48
7	5.6	5.33	5.40
8	5.6	5.53	5.58
9	5.4	5.83	5.64
10	6.5	5.67	5.68
11	5.1	5.80	5.56
12	5.8	5.30	5.72
13	5	5.67	5.54
14	6.2	5.60	5.86
15	5.6	6.17	5.74
16	6.7	5.83	5.84
17	5.2	5.80	5.76
18	5.5	5.50	5.66
19	5.8	5.47	5.48
20	5.1	5.57	5.78
21	5.8	5.87	5.72
22	6.7	5.90	5.76
23	5.2	5.97	5.90
24	6	5.67	
25	5.8		



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Example 3

Following data shows production volume (in '000 tones). Compute 4-year centered moving average

Year	Production(Y)	4-yearly moving average	4-year centered moving average
1995	21.0		
1996	22.0		
		22.75	
1997	23.0		23.13
		23.50	
1998	25.0		23.50
		23.50	
1999	24.0		23.75
		24.00	
2000	22.0		24.13
		24.25	
2001	25.0		24.63
		25.00	
2002	26.0		25.50
		26.00	
2003	27.0		26.38
		26.75	
2004	26.0		26.75
2005	28.0		

Weighted Moving Average

WEIGHTED MOVING AVERAGE (WMA)

Assigns decreasing weights to past data (Sum of weights = 1).

$$WMA_t = w_1Y_t + w_2Y_{t-1} + \ldots + w_kY_{t-k+1}$$

Methodology

Weight Selection

Weights (w_i) usually decrease for older data.
Example (k=3): 0.5, 0.3, 0.2 (sum = 1)

Forecast Rule

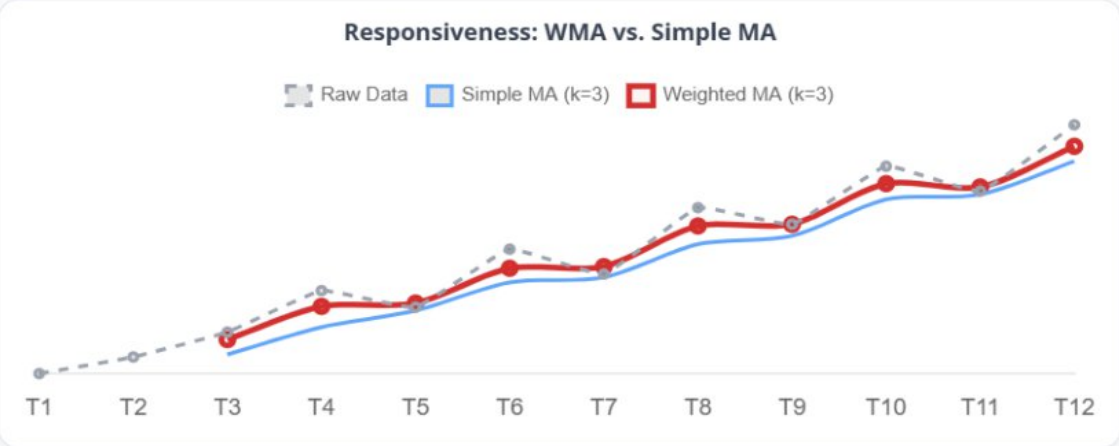
Next period's forecast is the current weighted average: $F_{t+1} = WMA_t$

Pros

- More responsive to recent changes.
- Better at capturing short-term trends.
- Prioritizes relevant recent data.

Cons

- Requires complex weight selection.
- Arbitrary weights can introduce bias.
- Still lags behind abrupt turning points.



Model Evaluation

To determine the best model (Simple vs Weighted MA), compare error metrics on a validation set:

MAPE
Mean Absolute Percentage Error
"Average % deviation"

RMSE
Root Mean Square Error
"Penalizes large errors"

→ Choose the model with the lower error score.

Practice Problem -1

Suppose we are given a time series data for 12 years -1989 to 2000 relating to sales of a certain business firm. These data are given below

Year	Sales (Million RS)
1989	10
1990	15
1991	20
1992	25
1993	15
1994	12
1995	15
1996	24
1997	15
1998	21
1999	15
2000	24

Find out the 3 year centered moving averages

Solution:

Year	Sales	Three year Moving Total	Three year MA
1989	10		
1990	15	45	15
1991	20	60	20
1992	25	60	20
1993	15	52	17
1994	12	42	14
1995	15	51	17
1996	24	54	18
1997	15	60	20
1998	21	54	18
1999	15	63	21
2000	24		

Practice problem-2

The following data represent 15 quarters of manufacturing capacity utilization

Quarter/Year	Utilization (%)
1/2003	82.50
2/2003	81.30
3/2003	81.30
4/2003	79.00
1/2004	76.60
2/2004	78.00
3/2004	78.40
4/2004	78.00
1/2005	78.80
2/2005	78.70
3/2005	78.40
4/2005	80.00
1/2006	80.70
2/2006	80.70
3/2006	80.80

Find the 2,3 & 4 quarter moving average

Thanks

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